

GSMA
Open Gateway

MENA Ignite Hackathon



GBP 10,000 PRIZE POOL

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Trusted Digital Identity & Cross-Border Verification

Sample Idea · TrustBridge - AI-Orchestrated Cross-Border Digital Identity & Verification Agent

Team Details

TEAM

TEAM NAME

[Enter your team name here]

TEAM MEMBERS

[First team member]

[Second team member]

[Third team member]

[Fourth team member]

[Fifth team member]

Integration Context

CONTEXT

Cross-border workers, migrants, and travellers across the MENA region routinely lose access to banking, telecom, and government services when they move between countries, because identity verification is fragmented, manual, and paper-based. Onboarding a migrant worker onto a new SIM, bank account, or e-government portal can take days and is highly vulnerable to identity fraud and SIM-swap-based account takeover. This problem is important because it directly blocks financial inclusion, delays remittances, and exposes both individuals and institutions to fraud losses at a regional scale.

TrustBridge uses CAMARA APIs available on the Nokia Network-as-Code platform like SIM Swap, Number Verification, and Device Status, as trusted, real-time signals that an AI agent layer orchestrates to verify a person's identity the moment they interact with a bank, telecom operator, airport kiosk, or government e-service in a new country. Instead of a user manually uploading documents and waiting for back-office checks, the agent silently confirms mobile ownership, checks for recent SIM-swap risk, and validates device legitimacy, then combines these signals into a single trust decision that downstream KYC/AML systems can rely on.

Theme Relevance

RELEVANCE

Millions of migrant workers, students, and travellers move across MENA borders every year and need to re-establish trusted digital identity quickly, to open a bank account, receive a salary, register a new SIM, or access a government portal. Manual, paper-heavy verification excludes people without strong documentation history and creates fraud opportunities (e.g., SIM-swap fraud used to hijack banking OTPs). TrustBridge turns telecom network intelligence into a portable, real-time trust signal that works the same way across borders, directly advancing regional financial inclusion and cross-border service continuity.

PROJECT TYPE

AI agent-orchestrated, telecom-signal-based cross-border digital identity verification platform.

GSMA PILLAR

Connectivity for Good

How it supports this pillar: TrustBridge treats mobile network connectivity as a portable trust anchor that travels with a person across borders. By having an AI agent continuously and autonomously reason over SIM Swap, Number Verification, and Device Status signals, rather than exposing these as one-off buttons a user must press, the solution converts raw network intelligence into safe, inclusive access to financial, telecom, and government services for people who would otherwise be excluded by slow manual verification.

Solution Overview

SOLUTION

TrustBridge is an AI agent that sits between a person's mobile number and any institution (bank, telco, airport authority, e-government portal) that needs to trust that person's identity in a new country. When a user initiates onboarding or a high-risk transaction, the agent autonomously decides which CAMARA signals to pull, reasons over the combined evidence, and returns a real-time Trust Score with an explainable rationale. TrustBridge is designed API-first so partner institutions can adopt it incrementally, calling the agent as a service rather than replacing their existing systems.

How it helps solve the problem: It collapses a multi-day, document-heavy cross-border verification process into a real-time decision, reduces SIM-swap-enabled account takeover fraud, and gives institutions a consistent, explainable trust signal for people without deep local credit or document history.

Key Features & Benefits

- Autonomous AI agent that decides which CAMARA APIs to call and when, based on transaction risk context.
- SIM Swap + Number Verification to detect account-takeover risk before approving high-value actions.
- Device Status checks to confirm device legitimacy and reachability at the point of onboarding.
- Explainable Trust Score with a visible agent reasoning trace for compliance officers.
- Cross-border profile portability, a traveller's trust signal is re-usable across partner institutions in different MENA countries.
- Graceful degradation with cached/sandbox fallback responses when a CAMARA API or model is rate-limited.
- Simple API-first integration so partner institutions can adopt the agent without replacing existing systems.

APIs & Technology

● APIs Used & How They Help

- **SIM Swap:** Confirms the SIM has not been recently swapped, flagging elevated account-takeover risk.
- **Number Verification:** Confirms mobile number ownership and matches it against the identity claim.
- **Device Status:** Confirms device legitimacy and reachability, adding an extra signal against spoofing.

● Tech Stack / Frameworks

- **Agent Orchestration:** LangGraph for stateful, multi-step orchestration of the verification workflow.
- **LLM:** Groq (Llama 3.3 70B, free tier) for fast reasoning; Google AI Studio (Gemini Flash) as a secondary model.
- **Memory:** Supabase pgvector for storing verification history and agent memory across sessions.
- **Frontend:** Next.js on Vercel for the institution-facing dashboard.
- **Data Source APIs:** CAMARA APIs (SIM Swap, Number Verification, Device Status) via Nokia Network-as-Code sandbox.
- **Database:** Supabase (Postgres) for audit trail, decisions, and compliance logging.

How the tech stack helps: LangGraph lets the agent maintain state across multi-step verification instead of a rigid script. Groq's free, high-rate-limit inference keeps the demo responsive, while Supabase pgvector gives the agent memory of prior verification attempts across borders, all using only tools from the approved Resource & Tooling Guide. All components are chosen from free or freemium tiers so the full prototype can be built, demoed, and iterated on without any paid commitment during the hackathon.

Innovation Highlights

INNOVATION

- ◆ Treats CAMARA telecom signals as autonomous agent tools rather than user-triggered actions.
- ◆ Portable cross-border trust score multiple institutions can consume without re-verifying from scratch.
- ◆ Visible reasoning trace showing why the agent approved, flagged, or escalated a case.
- ◆ Designed for graceful degradation with cached sandbox responses so live demos never fail on an API timeout.
- ◆ Built with a clear technical path from hackathon prototype to a pilot-ready, production-hardened service.

Impact Metrics

METRICS

◆ Reduction in average cross-border onboarding time (days to seconds).

◆ % reduction in SIM-swap-related fraud attempts detected and blocked.

◆ Number of institutions able to reuse a single trust score across a user's journey.

◆ Cost savings from reduced manual document review.

◆ Time for a new partner institution to integrate and go live with the agent.

◆

How the agents communicate

1 Client App → Orchestration Agent

User initiates onboarding or a high-risk transaction; request context is sent to the agent.

2 Orchestration Agent → CAMARA APIs (via Nokia NaC)

The agent autonomously decides which CAMARA APIs to invoke based on risk context.

3 Orchestration Agent → Trust Scoring Module

Combines API responses with prior memory into a weighted Trust Score and rationale.

4 Trust Scoring Module → Compliance Dashboard

Pushes the approval, flag, or escalation decision plus reasoning trace in real time.

Architecture components

● Input Layer

Web/mobile client used by the institution's front-line staff or end user.

● Orchestration Layer

LangGraph-based AI agent deciding which CAMARA tools to call and in what sequence.

● Verification Layer

Middleware invoking SIM Swap, Number Verification, and Device Status APIs.

● Risk & Trust Scoring Engine

Aggregates and weights signals plus agent memory into an explainable Trust Score.

● Dashboard / Agent UI

Displays decision, confidence, and the agent's reasoning trace.

● Audit & Compliance Module

Logs every API call, agent decision, and rationale for regulator review.

Important: Please insert or attach the links to your architecture diagrams in this document for your idea submission.

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READY TO BUILD?

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Smart Cities, Urban Safety & Mega-Project Infrastructure

Sample Idea • CityGuard - AI Agent for Real-Time Urban Safety & Lost-Person Response

Team Details

TEAM

TEAM NAME

[Enter your team name here]

TEAM MEMBERS

[First team member]

[Second team member]

[Third team member]

[Fourth team member]

[Fifth team member]

Integration Context

CONTEXT

In large MENA mega-projects and dense urban centres, families and event organisers have no fast, reliable way to locate a lost child, elderly relative, or worker in a crowded zone. Manual radio-based search is slow, and existing apps depend on the lost person actively using them, which fails exactly when it matters most. This is an important urban safety gap as MENA cities scale up giga-projects and mass gatherings.

CityGuard gives city operations centres and event organisers an AI agent that, once a guardian reports someone missing, autonomously uses Location Retrieval/Verification, Geofencing, and Device Status (via Nokia Network-as-Code) to locate the person's device, confirm it is reachable, and alert the nearest response team the moment the device enters a monitored zone.

Theme Relevance

RELEVANCE

MENA's mega-projects and giga-events bring together huge, unfamiliar crowds in newly built urban environments where getting separated from family or a work group is common. A network-based, AI-orchestrated response, rather than requiring the lost person to use an app, is directly relevant to keeping these new cities and events safe at scale.

PROJECT TYPE

AI-orchestrated urban safety and lost-person location agent for smart cities and mega-events.

GSMA PILLAR

Connectivity for Good

How it supports this pillar: CityGuard uses the mobile network itself, not a downloaded app, as the safety net for people in unfamiliar mega-city environments. By letting an AI agent autonomously combine Location, Geofencing, and Device Status signals, connectivity becomes a direct, life-relevant safety service rather than just a data channel.

Solution Overview

SOLUTION

CityGuard is an AI agent used by city operations or event safety teams. A guardian reports a missing person's phone number at a help desk or app; the agent autonomously retrieves location, sets a geofence around the last known area, and continuously checks device status, alerting the nearest response team in real time when the person is located. CityGuard is designed API-first so partner institutions can adopt it incrementally, calling the agent as a service rather than replacing their existing systems.

How it helps solve the problem: It reduces the time to locate a missing person in large, unfamiliar venues, coordinates response teams automatically instead of relying on manual radio calls, and works even if the missing person is not actively using a safety app themselves.

Key Features & Benefits

- AI agent that autonomously plans a search using location and geofencing signals.
- Real-time geofence alerts when the missing person's device enters a monitored zone.
- Device Status checks to confirm the phone is on and reachable before dispatching a team.
- Automatic routing of alerts to the nearest available response team.
- Works without requiring the missing person to have an app installed or actively open.
- Simple API-first integration so partner institutions can adopt the agent without replacing existing systems.

APIs & Technology

• APIs Used & How They Help

- **Location Retrieval / Verification:** Finds and confirms the approximate location of the reported device.
- **Geofencing:** Triggers an alert the moment the device enters or leaves a defined safety zone.
- **Device Status:** Confirms the device is reachable before a response team is dispatched.

• Tech Stack / Frameworks

- **Agent Orchestration:** LangGraph to manage the multi-step search-and-alert workflow.
- **LLM:** Google AI Studio (Gemini Flash) for fast reasoning over location updates.
- **Frontend:** Next.js on Vercel for the operations-centre map dashboard.
- **Memory:** Supabase pgvector to track active search cases and history.
- **Data Source APIs:** CAMARA APIs (Location, Geofencing, Device Status) via Nokia Network-as-Code.
- **Notifications:** Twilio for SMS/WhatsApp alerts to response teams and guardians.

How the tech stack helps: LangGraph keeps the search workflow stateful across multiple location checks over time, Gemini Flash's free tier keeps reasoning cost-free during the hackathon, and Twilio's trial credits let the demo show real alert delivery to a response team's phone. All components are chosen from free or freemium tiers so the full prototype can be built, demoed, and iterated on without any paid commitment during the hackathon.

Innovation Highlights

INNOVATION

- ◆ Turns geofencing into an active safety trigger instead of a passive analytics tool.
- ◆ No app install needed for the missing person, works purely off network signals.
- ◆ Automatically matches alerts to the nearest response team rather than a manual broadcast.
- ◆ Built with a clear technical path from hackathon prototype to a pilot-ready, production-hardened service.

Impact Metrics

METRICS

- ◆ Average time from report to location confirmed.
- ◆ % of cases resolved without manual radio search.
- ◆ Response team dispatch time after geofence alert.
- ◆ Time for a new partner institution to integrate and go live with the agent.

Methodology & Architecture

ARCHITECTURE

How the agents communicate

- 1 Help Desk / App → Safety Agent**
Guardian reports a missing person's mobile number and last known area.
- 2 Safety Agent → CAMARA APIs**
Agent retrieves location, sets a geofence, and checks device status.
- 3 Safety Agent → Response Team**
Sends a real-time alert with location once the device is found or the geofence triggers.

Architecture components

● Input Layer

Help desk kiosk or mobile app used by guardians and staff to file a report.

● Agent Layer

LangGraph-based search agent coordinating location and geofence checks.

● Verification Layer

Middleware invoking CAMARA Location, Geofencing, and Device Status APIs.

● Operations Dashboard

Live map view for the safety operations centre.

● Alerting Module

Sends real-time notifications to response teams and guardians.

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READY TO BUILD?

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Tourism, Pilgrimage & Cultural Experience Innovation

Sample Idea • PilgrimGuide - AI Family Reunification & Safety Copilot for Mass Pilgrimage

Team Details

TEAM

TEAM NAME

[Enter your team name here]

TEAM MEMBERS

[First team member]

[Second team member]

[Third team member]

[Fourth team member]

[Fifth team member]

Integration Context

CONTEXT

During Hajj, Umrah, and other mass religious gatherings, pilgrims often elderly, non-native-language speakers, and in unfamiliar surroundings, frequently become separated from their group in crowds of over a million people. Existing reunification processes rely on manual announcements and physical help desks, which do not scale to the density and multilingual nature of these events.

PilgrimGuide gives pilgrimage authorities an AI agent that, when a group leader reports a separated pilgrim, uses Location Retrieval/Verification and Device Reachability (via Nokia Network-as-Code) to find the pilgrim's approximate location and reachability, and coordinates the fastest reunification path, communicating with the pilgrim in their own language through a voice agent.

Theme Relevance

RELEVANCE

MENA hosts some of the largest religious and cultural gatherings in the world. Fast, dignified reunification for separated pilgrims, many of whom are elderly and non-Arabic-speaking, is a distinctly regional challenge that combines tourism experience with basic safety, making it a strong fit for network-intelligence-driven AI.

PROJECT TYPE

AI voice-enabled family reunification and safety copilot for mass pilgrimage events.

GSMA PILLAR

Connectivity for Good

How it supports this pillar: PilgrimGuide uses mobile connectivity to protect the dignity and safety of pilgrims during the most vulnerable moment of their journey, being lost in an unfamiliar, enormous crowd. By having an AI agent reason over location and reachability signals and speak to the pilgrim in their own language, the project extends the network's value beyond connectivity into direct human care.³ Idea and Approach Details

Solution Overview

SOLUTION

PilgrimGuide is an AI agent used at pilgrimage help centres. A group leader reports a separated pilgrim's phone number; the agent retrieves an approximate location, confirms the device is reachable, and places a voice call in the pilgrim's language to calmly guide them to the nearest help point, while directing a volunteer team to the same location. PilgrimGuide is designed API-first so partner institutions can adopt it incrementally, calling the agent as a service rather than replacing their existing systems.

How it helps solve the problem: It cuts the time to reunite separated pilgrims with their group, removes language as a barrier during a stressful moment, and reduces the burden on manual help-desk staff during peak crowd periods.

Key Features & Benefits

- AI voice agent that speaks with the separated pilgrim in their own language.
- Location Retrieval to identify the pilgrim's approximate position in a massive crowd.
- Device Reachability checks to confirm the agent can actually reach the pilgrim's phone.
- Automatic dispatch of the nearest volunteer or help-desk team to the same area.
- Works even for pilgrims unfamiliar with smartphone apps, since it uses a simple voice call.
- Simple API-first integration so partner institutions can adopt the agent without replacing existing systems.

APIs & Technology

● APIs Used & How They Help

- **Location Retrieval / Verification:** Identifies the approximate location of the separated pilgrim's device.
- **Device Reachability:** Confirms the device can be reached before initiating the voice call.

● Tech Stack / Frameworks

- **Agent Orchestration:** CrewAI with a 'Reunification Coordinator' agent role.
- **LLM:** Google AI Studio (Gemini Flash) for multilingual reasoning and conversation planning.
- **Voice:** ElevenLabs for natural multilingual text-to-speech; Deepgram for speech-to-text.
- **Telephony:** Vapi to place and manage the AI phone call to the pilgrim.
- **Data Source APIs:** CAMARA APIs (Location, Device Reachability) via Nokia Network-as-Code.

How the tech stack helps: Vapi and ElevenLabs let the team demo a real, natural-sounding multilingual phone call within the hackathon's time budget, while Gemini Flash's free tier handles the conversational reasoning needed to calmly guide a distressed pilgrim to safety. All components are chosen from free or freemium tiers so the full prototype can be built, demoed, and iterated on without any paid commitment during the hackathon.

Innovation Highlights

INNOVATION

- ◆ Combines a voice-first AI agent with telecom location signals for a non-app-dependent safety tool.
- ◆ Designed around dignity and calm guidance for elderly or distressed pilgrims, not just raw geolocation.
- ◆ Coordinates human volunteers and AI guidance together rather than replacing one with the other.
- ◆ Built with a clear technical path from hackathon prototype to a pilot-ready, production-hardened service.

Impact Metrics

METRICS

- ◆ Average time to reunite a separated pilgrim with their group.
- ◆ Number of languages successfully handled by the voice agent.
- ◆ Reduction in manual help-desk workload during peak hours.
- ◆ Time for a new partner institution to integrate and go live with the agent.

Methodology & Architecture

ARCHITECTURE

How the agents communicate

- 1 Group Leader / Help Desk → Reunification Agent**
Reports a separated pilgrim's phone number and last known group location.
- 2 Reunification Agent → CAMARA APIs**
Retrieves location and confirms device reachability.
- 3 Reunification Agent → Pilgrim (Voice Call) & Volunteer Team**
Guides the pilgrim by voice while directing volunteers to the same area.

Architecture components

● Input Layer

Help-desk console used by group leaders and pilgrimage staff.

● Agent Layer

CrewAI-based reunification coordinator managing the search-and-guide workflow.

● Verification Layer

Middleware invoking Location and Device Reachability APIs.

● Voice Interface

AI phone call system guiding the pilgrim in their own language.

● Volunteer Dispatch

Notifies the nearest available volunteer team.

Important: Please insert or attach the links to your architecture diagrams in this document for your idea submission.

Diagram Link: *[Insert Lucidchart / Figma / Miro / Excalidraw link here]*



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READY TO BUILD?

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Secure Fintech, Payments & Anti-Fraud Innovation

Sample Idea • PayShield AI - Real-Time Card-Not-Present Fraud Prevention Agent

Team Details

TEAM

TEAM NAME

[Enter your team name here]

TEAM MEMBERS

[First team member]

[Second team member]

[Third team member]

[Fourth team member]

[Fifth team member]

Integration Context

CONTEXT

Online and card-not-present payment fraud is rising sharply across MENA e-commerce and digital banking, particularly through SIM-swap-enabled OTP interception. Existing fraud rules are static and slow to adapt, causing both missed fraud and unnecessary declines on legitimate transactions, which frustrates genuine customers.

PayShield AI gives payment gateways and digital banks an AI agent that, for every card-not-present transaction above a configurable risk threshold, autonomously checks SIM Swap, Number Verification, and Device Status (via Nokia Network-as-Code) in real time before authorising the payment, rather than relying purely on static rule engines.

Theme Relevance

RELEVANCE

Digital payments adoption is accelerating across MENA, and fraud losses directly undermine trust in fintech and e-commerce growth. A telecom-signal-based, AI-orchestrated fraud check that runs in milliseconds during checkout is directly relevant to securing the region's fast-growing digital payments ecosystem.

PROJECT TYPE

AI agent-orchestrated, real-time payment fraud-prevention layer for fintechs and payment gateways.

GSMA PILLAR

Connectivity for Good

How it supports this pillar: PayShield AI turns telecom network signals into a real-time trust layer for digital payments, protecting both consumers and merchants from fraud without adding friction for legitimate customers. This is a direct example of mobile connectivity being used for good in everyday financial life.³ Idea and Approach Details

Solution Overview

SOLUTION

PayShield AI plugs into a payment gateway's checkout flow. For flagged transactions, an AI agent autonomously decides which CAMARA checks to run, combines the results with transaction context (amount, merchant risk, time), and returns an approve/step-up/decline recommendation with an explainable rationale, all within the checkout's latency budget. PayShield AI is designed API-first so partner institutions can adopt it incrementally, calling the agent as a service rather than replacing their existing systems.

How it helps solve the problem: It reduces fraud losses from SIM-swap-enabled OTP interception, cuts false declines on legitimate transactions by adding a genuine identity signal rather than relying on rules alone, and gives fraud teams an explainable audit trail for every decision.

Key Features & Benefits

- Real-time AI agent decisioning integrated directly into the payment checkout flow.
- SIM Swap check to catch OTP-interception fraud before authorisation.
- Number Verification and Device Status as additional identity corroboration signals.
- Step-up authentication recommendation for medium-risk cases instead of an outright decline.
- Explainable decision log for fraud and compliance teams.
- Simple API-first integration so partner institutions can adopt the agent without replacing existing systems.

APIs & Technology

APIs Used & How They Help

- **SIM Swap:** Detects a recent SIM swap that commonly precedes OTP-interception fraud.
- **Number Verification:** Confirms the phone number used for OTP matches the account holder's registered number.
- **Device Status:** Confirms the device used for the transaction is legitimate and reachable.

Tech Stack / Frameworks

- **Agent Orchestration:** Pydantic AI for a type-safe, production-reliable decisioning agent.
- **LLM:** Groq (Llama 3.3 70B) for low-latency reasoning suited to checkout-time decisions.
- **Backend:** n8n workflow automation to wire the agent into an existing payment gateway webhook.
- **Database:** Neon serverless Postgres for transaction and decision logging.
- **Data Source APIs:** CAMARA APIs (SIM Swap, Number Verification, Device Status) via Nokia Network-as-Code.

How the tech stack helps: Pydantic AI's type safety reduces the risk of malformed decisions in a payments context, Groq keeps latency low enough for a live checkout demo, and n8n makes it fast to simulate a realistic payment gateway webhook integration during the hackathon. All components are chosen from free or freemium tiers so the full prototype can be built, demoed, and iterated on without any paid commitment during the hackathon.

Innovation Highlights

INNOVATION

- ◆ Uses SIM Swap specifically to counter OTP-interception fraud, a fast-growing regional fraud pattern.
- ◆ Recommends step-up authentication instead of a binary approve/decline, reducing false declines.
- ◆ Designed to plug into existing payment gateways via a webhook rather than requiring a rebuild.
- ◆ Built with a clear technical path from hackathon prototype to a pilot-ready, production-hardened service.

Impact Metrics

METRICS

- ◆ % reduction in confirmed fraud losses on flagged transactions.
- ◆ % reduction in false declines on legitimate transactions.
- ◆ Average added latency per transaction from the fraud check.
- ◆ Time for a new partner institution to integrate and go live with the agent.

Methodology & Architecture

ARCHITECTURE

How the agents communicate

- 1 Checkout → Fraud Agent**
Payment gateway sends a webhook for transactions above the risk threshold.
- 2 Fraud Agent → CAMARA APIs**
Agent autonomously calls SIM Swap, Number Verification, and Device Status.
- 3 Fraud Agent → Payment Gateway**
Returns approve, step-up, or decline recommendation with rationale.

Architecture components

● Input Layer

Payment gateway checkout webhook triggering the fraud check.

● Agent Layer

Pydantic AI decisioning agent orchestrating CAMARA signal checks.

● Verification Layer

Middleware invoking SIM Swap, Number Verification, and Device Status APIs.

● Decision Log

Stores every transaction's risk signals and final decision for audit.

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READY TO BUILD?

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Industrial & Enterprise AI Automation

Sample Idea • FactoryPulse - AI Predictive Maintenance Agent for Connected Factory Floors

Team Details

TEAM

TEAM NAME

[Enter your team name here]

TEAM MEMBERS

[First team member]

[Second team member]

[Third team member]

[Fourth team member]

[Fifth team member]

Integration Context

CONTEXT

Smart factories across the region increasingly rely on connected sensors and robotic systems, but unplanned downtime from undetected connectivity degradation (a robot's control link losing quality moments before a fault) still causes costly production stops. Maintenance teams typically find out only after a failure has already happened.

FactoryPulse gives plant managers an AI agent that continuously monitors Network Intelligence and Device Status signals (via Nokia Network-as-Code) for critical robotic and IoT equipment, autonomously requesting QoS on Demand for at-risk devices and alerting maintenance teams before a connectivity-related failure causes downtime.

Theme Relevance

RELEVANCE

Industrial automation and smart manufacturing are strategic priorities for MENA economies diversifying beyond oil and gas. Preventing costly downtime by treating network quality as an early-warning signal for equipment health is directly relevant to accelerating trustworthy industrial AI adoption in the region.

PROJECT TYPE

AI predictive-maintenance agent using network intelligence as an early-warning signal for industrial equipment.

GSMA PILLAR

Connectivity for Good

How it supports this pillar: FactoryPulse shows that network quality itself is valuable operational data, not just a pipe for data to flow through. By having an AI agent proactively manage quality of service for critical machinery, the project turns telecom intelligence into direct industrial safety and productivity gains.³ Idea and Approach Details

Solution Overview

SOLUTION

FactoryPulse is an AI agent that plant managers connect to their critical robotic and IoT devices. The agent continuously watches network intelligence and device status signals for early signs of degradation, autonomously requests a QoS boost for at-risk devices, and raises a maintenance ticket with a specific, explainable reason before a failure occurs. FactoryPulse is designed API-first so partner institutions can adopt it incrementally, calling the agent as a service rather than replacing their existing systems.

How it helps solve the problem: It shifts maintenance from reactive to predictive by treating connectivity health as an early-warning signal, reduces unplanned downtime, and prioritises limited network capacity for the machines that need it most at any given moment.

Key Features & Benefits

- Continuous AI monitoring of network quality and device status for critical equipment.
- Autonomous QoS on Demand requests for machines showing early signs of degradation.
- Predictive maintenance tickets with an explainable, signal-based rationale.
- Prioritisation logic so QoS boosts go to the most production-critical machines first.
- Simple API-first integration so partner institutions can adopt the agent without replacing existing systems.

APIs & Technology

● APIs Used & How They Help

- **Network Intelligence:** Provides early signals of connectivity degradation for connected machinery.
- **Device Status:** Confirms whether a specific device is online, reachable, and behaving normally.
- **QoS on Demand:** Requests guaranteed network quality for machines identified as at-risk.

● Tech Stack / Frameworks

- **Agent Orchestration:** LangGraph to manage continuous monitoring and escalation as a long-running workflow.
- **LLM:** Cerebras Inference for ultra-fast analysis of streaming network signals.
- **Frontend:** Next.js on Vercel for the plant manager's monitoring dashboard.
- **Database:** Upstash Redis for fast, low-latency storage of streaming device signals.
- **Data Source APIs:** CAMARA APIs (Network Intelligence, Device Status, QoS on Demand) via Nokia Network-as-Code.

How the tech stack helps: LangGraph's stateful graph is well suited to a long-running monitor-and-escalate loop, Cerebras's free developer tier gives very fast inference for near-real-time signal analysis, and Upstash Redis handles the high-frequency signal updates a factory floor produces without added infrastructure cost. All components are chosen from free or freemium tiers so the full prototype can be built, demoed, and iterated on without any paid commitment during the hackathon.

Innovation Highlights

INNOVATION

- ◆ Uses telecom network intelligence as a predictive maintenance signal, not just a connectivity metric.
- ◆ Prioritises QoS on Demand dynamically based on which machine is most production-critical right now.
- ◆ Moves maintenance decisions from reactive fault response to proactive, explainable prediction.
- ◆ Built with a clear technical path from hackathon prototype to a pilot-ready, production-hardened service.

Impact Metrics

METRICS

- ◆ Reduction in unplanned downtime hours per month.
- ◆ Number of failures pre-empted via early QoS intervention.
- ◆ Maintenance cost savings from predictive vs. reactive repairs.
- ◆ Time for a new partner institution to integrate and go live with the agent.

Methodology & Architecture

ARCHITECTURE

How the agents communicate

- 1 Factory IoT Devices → FactoryPulse Agent**
Continuously stream network and status signals to the agent.
- 2 FactoryPulse Agent → CAMARA APIs**
Analyses network intelligence and requests QoS on Demand for at-risk devices.
- 3 FactoryPulse Agent → Maintenance Team**
Raises a predictive maintenance ticket with a clear, signal-based rationale.

Architecture components

● Input Layer

Connected robotic and IoT devices on the factory floor.

● Agent Layer

LangGraph-based monitoring agent analysing streaming signals.

● Verification Layer

Middleware invoking Network Intelligence, Device Status, and QoS on Demand APIs.

● Maintenance Dashboard

Displays at-risk equipment and predictive maintenance tickets.

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READY TO BUILD?

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Climate Resilience & Environmental Monitoring

Sample Idea • ClimaWatch - AI Early-Warning Agent for Flash Floods & Extreme Heat

Team Details

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[Enter your team name here]

TEAM MEMBERS

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Integration Context

CONTEXT

MENA cities are increasingly exposed to flash floods and extreme heat events, but early-warning systems often depend on fixed weather stations with sparse coverage, missing hyper-local conditions in vulnerable neighbourhoods and along drainage-prone road networks.

ClimaWatch gives municipal disaster-management teams an AI agent that aggregates signals from distributed IoT environmental sensors, using Device Status to confirm sensor health and QoS on Demand to guarantee data delivery from sensors during the exact extreme-weather window when networks are most congested, with Location used to pinpoint affected zones.

Theme Relevance

RELEVANCE

Climate resilience is an urgent, region-wide priority as MENA faces more frequent flash floods and extreme heat. A network-intelligence-driven early-warning system that keeps working precisely when infrastructure is under the most stress is directly relevant to protecting vulnerable communities.

PROJECT TYPE

AI-orchestrated early-warning agent for hyper-local flood and heat events using IoT sensor networks.

GSMA PILLAR

Connectivity for Good

How it supports this pillar: ClimaWatch ensures that life-saving environmental data keeps flowing exactly when a network is most likely to fail, during an extreme weather event, by having an AI agent proactively guarantee quality of service for critical sensors. This is a direct, high-stakes example of connectivity being used for good.

ClimaWatch is an AI agent connected to a network of low-cost IoT environmental sensors (rainfall, temperature, water level) placed across a city. During a developing extreme-weather event, the agent checks sensor device status, requests QoS on Demand to keep the most critical sensors reporting, and pinpoints affected zones by location, issuing an early warning to municipal teams and residents. ClimaWatch is designed API-first so partner institutions can adopt it incrementally, calling the agent as a service rather than replacing their existing systems.

How it helps solve the problem: It extends effective monitoring coverage using low-cost distributed sensors instead of relying only on sparse fixed weather stations, and ensures the data keeps flowing during the exact moments, storms, heatwaves, when networks are most congested and most likely to drop critical readings.

Key Features & Benefits

- AI agent aggregating distributed low-cost sensor data into a single early-warning signal.
- QoS on Demand requested automatically for critical sensors during a developing extreme event.
- Device Status checks to detect and route around failed or unreachable sensors.
- Location-based zoning to issue precise, neighbourhood-level warnings rather than city-wide alerts.
- Simple API-first integration so partner institutions can adopt the agent without replacing existing systems.

APIs & Technology

● APIs Used & How They Help

- **Device Status:** Confirms which environmental sensors are online and reporting reliably.
- **QoS on Demand:** Guarantees data delivery from critical sensors during high-network-congestion extreme weather events.
- **Location Verification:** Pinpoints which neighbourhoods or zones are affected for targeted warnings.

● Tech Stack / Frameworks

- **Agent Orchestration:** LangGraph to manage the continuous monitoring and escalation-to-warning workflow.
- **LLM:** Google AI Studio (Gemini Flash) for interpreting sensor trends and drafting warning messages.
- **Frontend:** Streamlit Community Cloud for the municipal operations dashboard.
- **Memory:** Chroma for storing historical sensor patterns used to detect anomalies.
- **Data Source APIs:** CAMARA APIs (Device Status, QoS on Demand, Location) via Nokia Network-as-Code.
- **Alerts:** Twilio for SMS-based public warnings to residents in affected zones.

How the tech stack helps: LangGraph naturally models the escalation path from raw sensor readings to a confirmed warning, Gemini Flash helps translate technical readings into clear public warning language, and Twilio's trial credits allow a real SMS alert demo to affected residents during judging. All components are chosen from free or freemium tiers so the full prototype can be built, demoed, and iterated on without any paid commitment during the hackathon.

Innovation Highlights

INNOVATION

- ◆ Uses QoS on Demand specifically to protect critical sensor data during the network stress of an extreme event.
- ◆ Combines low-cost distributed IoT sensors with telecom signals rather than relying solely on expensive fixed stations.
- ◆ Issues precise, zone-level public warnings instead of broad city-wide alerts.
- ◆ Built with a clear technical path from hackathon prototype to a pilot-ready, production-hardened service.

Impact Metrics

METRICS

- ◆ Lead time gained before flood/heat impact vs. existing warning systems.
- ◆ % of critical sensor readings successfully delivered during peak network congestion.
- ◆ Number of residents reached with zone-specific warnings.
- ◆ Time for a new partner institution to integrate and go live with the agent.

Methodology & Architecture

ARCHITECTURE

How the agents communicate

- 1 IoT Sensors → ClimaWatch Agent**
Continuously report rainfall, temperature, and water-level readings.
- 2 ClimaWatch Agent → CAMARA APIs**
Checks device status, requests QoS for critical sensors, and verifies affected locations.
- 3 ClimaWatch Agent → Municipal Team & Residents**
Issues an early warning with zone-specific detail.

Architecture components

● Input Layer

Distributed low-cost IoT environmental sensors across the city.

● Agent Layer

LangGraph-based monitoring and early-warning agent.

● Verification Layer

Middleware invoking Device Status, QoS on Demand, and Location APIs.

● Alerting Module

Sends SMS warnings to municipal teams and residents in affected zones.

Important: Please insert or attach the links to your architecture diagrams in this document for your idea submission.

Diagram Link: [Insert Lucidchart / Figma / Miro / Excalidraw link here]



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Sample Idea • HealthLink AI - Trusted Telemedicine Identity & Reliability Copilot

Team Details

TEAM

TEAM NAME

[Enter your team name here]

TEAM MEMBERS

[First team member]

[Second team member]

[Third team member]

[Fourth team member]

[Fifth team member]

Integration Context

CONTEXT

Telemedicine adoption in remote and underserved MENA communities is held back by two issues: patients' identities are hard to verify remotely for prescription and insurance purposes, and video consultations frequently degrade or drop in areas with inconsistent connectivity, undermining trust in remote care.

HealthLink AI gives telemedicine platforms an AI agent that verifies a patient's identity using Number Verification and SIM Swap (via Nokia Network-as-Code) before a prescription or insurance claim is processed, and requests QoS on Demand to guarantee call quality for the duration of the consultation, especially in low-infrastructure rural areas.

Theme Relevance

RELEVANCE

This idea spans multiple hackathon themes, trusted identity, fintech-adjacent insurance claims, and connectivity-guaranteed service delivery, making it a strong fit for Open Innovation. Reliable, trustworthy telemedicine is a pressing need for MENA's underserved rural populations who lack easy access to physical clinics.

PROJECT TYPE

AI copilot combining identity verification and guaranteed connectivity for trusted telemedicine.

GSMA PILLAR

Connectivity for Good

How it supports this pillar: HealthLink AI uses telecom signals to solve two barriers to healthcare access at once: verifying who a remote patient really is, and guaranteeing the call quality needed for a doctor to safely diagnose them. This is a clear case of mobile connectivity being used for good in an essential public service.³

Idea and Approach Details

Solution Overview

SOLUTION

HealthLink AI is an AI agent embedded in a telemedicine platform. Before a consultation begins, it verifies the patient's identity for prescription/insurance purposes using Number Verification and SIM Swap checks, then autonomously requests a QoS boost for the video/voice call, monitoring quality throughout and alerting the platform if the connection risks dropping. HealthLink AI is designed API-first so partner institutions can adopt it incrementally, calling the agent as a service rather than replacing their existing systems.

How it helps solve the problem: It builds trust in remote prescriptions and insurance claims by tying them to a verified identity, and directly improves clinical safety and patient experience by proactively protecting call quality instead of letting consultations degrade or drop.

Key Features & Benefits

- Pre-consultation identity verification for prescription and insurance validity.
- SIM Swap check to prevent identity fraud in remote prescription requests.
- Automatic QoS on Demand request for the full duration of each video consultation.
- Real-time call-quality monitoring with proactive alerts if a drop is imminent.
- Simple API-first integration so partner institutions can adopt the agent without replacing existing systems.

APIs & Technology

● APIs Used & How They Help

- **Number Verification:** Confirms the patient's mobile number matches their registered identity.
- **SIM Swap:** Flags a recently swapped SIM as a fraud risk before approving a prescription or claim.
- **QoS on Demand:** Guarantees call quality for the full duration of the telemedicine consultation.

● Tech Stack / Frameworks

- **Agent Orchestration:** CrewAI with separate 'Identity Verifier' and 'Call Quality Monitor' agent roles.
- **LLM:** Google AI Studio (Gemini Flash) for reasoning over combined identity and quality signals.
- **Frontend:** Next.js on Vercel for the clinician-facing dashboard.
- **Database:** Supabase for patient verification and consultation quality logs.
- **Data Source APIs:** CAMARA APIs (Number Verification, SIM Swap, QoS on Demand) via Nokia Network-as-Code.

How the tech stack helps: CrewAI's role-based structure cleanly separates the identity-verification concern from the ongoing call-quality-monitoring concern, while Gemini Flash's free tier supports both reasoning tasks without added cost during the hackathon build. All components are chosen from free or freemium tiers so the full prototype can be built, demoed, and iterated on without any paid commitment during the hackathon.

Innovation Highlights

INNOVATION

- ◆ Combines identity trust and connectivity guarantee in a single copilot for telemedicine, spanning multiple themes.
- ◆ Uses SIM Swap specifically to prevent remote-prescription identity fraud, an underserved fraud vector.
- ◆ Proactively protects call quality rather than reacting only after a consultation has already degraded.
- ◆ Built with a clear technical path from hackathon prototype to a pilot-ready, production-hardened service.

Impact Metrics

METRICS

◆ % reduction in dropped or degraded telemedicine consultations.

◆ % reduction in fraudulent prescription/insurance claims linked to identity fraud.

◆ Patient and clinician satisfaction scores for remote consultations.

◆ Time for a new partner institution to integrate and go live with the agent.

Methodology & Architecture

ARCHITECTURE

How the agents communicate

- 1 Telemedicine Platform → HealthLink Agent**
Requests identity verification before a consultation begins.
- 2 HealthLink Agent → CAMARA APIs**
Checks Number Verification, SIM Swap, and requests QoS for the call.
- 3 HealthLink Agent → Platform / Clinician**
Confirms identity status and proactively alerts on call-quality risk.

Architecture components

● Input Layer

Telemedicine platform's consultation booking and video call flow.

● Agent Layer

CrewAI-based identity and call-quality agents.

● Verification Layer

Middleware invoking Number Verification, SIM Swap, and QoS on Demand APIs.

● Compliance Log

Stores identity verification and call-quality records for audits.

Important: Please insert or attach the links to your architecture diagrams in this document for your idea submission.

Diagram Link: [Insert Lucidchart / Figma / Miro / Excalidraw link here]



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